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**INSTITUTE**

The diagram consists of a rectangular box at the top with the word 'INSTITUTE' centered inside. A large, hollow arrow points downwards from the bottom center of this box to a banner below. The banner is a horizontal rectangle with wavy, ribbon-like edges, containing the words 'VISION' and 'MISSION' separated by a space. The entire diagram is enclosed within a double-line border.

**VISION MISSION**

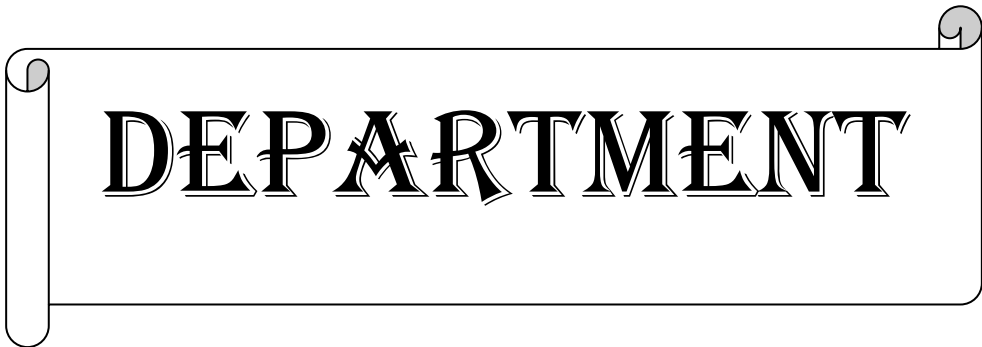
# INSTITUTE VISION AND MISSION

## VISION

To be a premier technological institute striving for excellence with global perspective and commitment to the nation.

## MISSION

- To produce engineering graduates of professional quality and global perspective through Learner Centric Education.
- To establish linkages with government, industry and research laboratories to promote R&D activities and to disseminate innovations.
- To create an eco-system in the institute that leads to holistic development and ability for life-long learning..



**DEPARTMENT**



**VISION MISSION**



## **DEPARTMENT VISION AND MISSION**

### **Vision:**

- To evolve as a centre of academic and research excellence in the area of Computer Science and Engineering.

### **Mission :**

- To utilize innovative learning methods for academic improvement.
- To encourage higher studies and research to meet the futuristic requirements of Computer Science and Engineering.
- To inculcate Ethics and Human values for developing students with good character



**PROGRAM  
EDUCATIONAL  
OBJECTIVES,  
PROGRAM OUTCOMES  
& PROGRAM  
SPECIFIC  
OUTCOMES**

## **Program Educational Objectives (PEOs)**

Graduates of this programme will :

**PEO 1:** Adapt to evolving technology.

**PEO 2:** Provide optimal solutions to real time problems.

**PEO 3:** Demonstrate his/her abilities to support service activities with due consideration for Professional and Ethical Values.

## **Programme Specific Outcomes (PSO s):**

A graduate of the Computer Science and Engineering Program will be able to:

**PSO 1:** Use Mathematical Abstractions and Algorithmic Design along with Open Source Programming tools to solve complexities involved in Programming. [K3]

**PSO 2:** Use Professional engineering practices and strategies for development and maintenance of software. [K3]

# Program Outcomes (POs):

## Computer Science Engineering Graduates will be able to:

1. **Engineering knowledge:** Apply the knowledge of Mathematics, Science, Engineering Fundamentals and Concepts of Computer Science Engineering to the solution of complex Engineering problems. [K3]
2. **Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of Mathematics, Natural Sciences and Computer Science. [K4]
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specific needs with appropriate consideration for the public health and safety, and the cultural, societal and environmental considerations. [K5]
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions. [K5]
5. **Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex Engineering activities with an understanding of the limitations. [K3]
6. **The Engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional Engineering practice. [K3]
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development. [K3]
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the Engineering practice. [K3]
9. **Individual and team work:** Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings. [K6]
10. **Communication:** Communicate effectively on complex Engineering activities with the Engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions. [K2]
11. **Project management and finance:** Demonstrate knowledge and understanding of the Engineering and Management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments. [K6]
12. **Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change. [K1]

# **ACADEMIC CALENDAR**

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**Directorate of Academic Planning**  
JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA  
KAKINADA-533003, Andhra Pradesh, INDIA  
(Established by AP Government Act No. 30 of 2008)

*Ln. No. 01-08/JNTUK/DAP/AC/B. Tech-B. Pharmacy/II-III-IV Year/2020-21*

*Date: 29-12-2020*

**Dr. R. Srinivasa Rao,**  
Director, Academic Planning  
JNTUK, Kakinada

To  
All the Principals of Affiliated Colleges,  
JNTUK, Kakinada.

## **Academic Calendar for II, III and IV - B. Tech & B. Pharmacy Academic year 2020-21**

| <b>I SEMESTER</b>                                                                     |            |            |       |
|---------------------------------------------------------------------------------------|------------|------------|-------|
| Description                                                                           | From       | To         | Weeks |
| Commencement of Class Work                                                            | 02.11.2020 |            |       |
| I Unit of Instruction                                                                 | 02.11.2020 | 19.12.2020 | 7W    |
| II Unit of Instructions                                                               | 21.12.2020 | 23.01.2021 | 5W    |
| I Mid Examinations                                                                    | 25.01.2021 | 30.01.2021 | 1W    |
| II Unit of Instructions(Continued)                                                    | 01.02.2021 | 20.02.2021 | 3W    |
| II Mid Examinations                                                                   | 22.02.2021 | 27.02.2021 | 1W    |
| Preparation & Practicals                                                              | 01.03.2021 | 06.03.2021 | 1W    |
| End Examinations                                                                      | 08.03.2021 | 20.03.2021 | 2W    |
| Commencement of II Semester Class Work                                                | 22.03.2021 |            |       |
| <b>II SEMESTER</b>                                                                    |            |            |       |
| I Unit of Instructions                                                                | 22.03.2021 | 08.05.2021 | 7W    |
| I Mid Examinations                                                                    | 10.05.2021 | 12.05.2021 | 1/2W  |
| II Unit of Instructions                                                               | 13.05.2021 | 30.06.2021 | 7W    |
| II Mid Examinations                                                                   | 01.07.2021 | 03.07.2021 | 1/2W  |
| Preparation & Practicals                                                              | 05.07.2021 | 10.07.2021 | 1W    |
| End Examinations                                                                      | 12.07.2021 | 24.07.2021 | 2W    |
| Commencement of next Year Class Work                                                  |            |            |       |
| <i>Note: Calendar is prepared with 8 hrs/day hence 7 weeks per instruction period</i> |            |            |       |

*R. Srinivasa Rao*  
Director Academic Planning  
Academic Planning  
JNTUK Kakinada

Copy to the Secretary to the Hon'ble Vice Chancellor, JNTUK  
Copy to Rector, JNTUK  
Copy to Registrar, JNTUK  
Copy to Director Academic Audit, JNTUK  
Copy to Director of Evaluation, JNTUK



# SRI VASAVI ENGINEERING COLLEGE (Autonomous)

Pedatadepalli, TADEPALLIGUDEM-534 101, W.G. Dist.

Department Of Computer Science & Engineering (Accredited by NBA)



## CLASS CONSOLIDATED TIME TABLE

Class: IV B.Tech II Semester

Sec : A

Class Coordinator :Mr.M.Baburao

w.e.f. 31-03-2021

Room No :G-301

| Periods     | 1                       | 2                       | 3                       | 4                       | 1:00PM<br>2:00PM | 5                       | 6                       | 7                       |
|-------------|-------------------------|-------------------------|-------------------------|-------------------------|------------------|-------------------------|-------------------------|-------------------------|
| Time<br>Day | (09.30 AM-<br>10.30 AM) | (10.30 AM-<br>11.20 AM) | (11.20 AM-<br>12.10 PM) | (12.10 PM-<br>01.00 PM) |                  | (02.00 PM-<br>02.50 PM) | (02.50 PM-<br>03.40 PM) | (03.40 PM-<br>04.30 PM) |
| Mon         | DS                      | MS                      | MS                      | CPP                     | LUNCH<br>BREAK   | ML                      | DS                      | LIB                     |
| Tue         | CPP                     | ML                      | DS                      | SEMINAR                 |                  | CPP                     | MS                      | SPORTS                  |
| Wed         | MS                      | CPP                     | ML                      | DS                      |                  | PROJECT LAB             |                         |                         |
| Thu         | ML                      | CPP                     | SEMINAR                 |                         |                  | DS                      | MS                      | ML                      |

Class: IV B.Tech II Semester

Sec : B

Class Coordinator :Mrs.D.S.L.Manikateswari

w.e.f. 31-03-2021

Room No :G-302

| Periods     | 1                       | 2                       | 3                       | 4                       | 1:00PM<br>2:00PM | 5                       | 6                       | 7                       |
|-------------|-------------------------|-------------------------|-------------------------|-------------------------|------------------|-------------------------|-------------------------|-------------------------|
| Time<br>Day | (09.30 AM-<br>10.30 AM) | (10.30 AM-<br>11.20 AM) | (11.20 AM-<br>12.10 PM) | (12.10 PM-<br>01.00 PM) |                  | (02.00 PM-<br>02.50 PM) | (02.50 PM-<br>03.40 PM) | (03.40 PM-<br>04.30 PM) |
| Mon         | MS                      | CPP                     | ML                      | DS                      | LUNCH<br>BREAK   | PROJECT LAB             |                         |                         |
| Tue         | DS                      | MS                      | MS                      | CPP                     |                  | ML                      | DS                      | ML                      |
| Wed         | MS                      | SEMINAR                 |                         | CPP                     |                  | ML                      | DS                      | SEMINAR                 |
| Thu         | CPP                     | DS                      | ML                      | LIB                     |                  | MS                      | CPP                     | SPORTS                  |

Class: IV B.Tech II Semester

Sec : C

Class Coordinator :Ms.D.SasiRekha

w.e.f. 31-03-2021

Room No : G-401

| Periods     | 1                       | 2                       | 3                       | 4                       | 1:00PM<br>2:00PM | 5                       | 6                       | 7                       |
|-------------|-------------------------|-------------------------|-------------------------|-------------------------|------------------|-------------------------|-------------------------|-------------------------|
| Time<br>Day | (09.30 AM-<br>10.30 AM) | (10.30 AM-<br>11.20 AM) | (11.20 AM-<br>12.10 PM) | (12.10 PM-<br>01.00 PM) |                  | (02.00 PM-<br>02.50 PM) | (02.50 PM-<br>03.40 PM) | (03.40 PM-<br>04.30 PM) |
| Mon         | CPP                     | ML                      | CPP                     | MS                      | LUNCH<br>BREAK   | ML                      | DS                      | SPORTS                  |
| Tue         | DS                      | SEMINAR                 | MS                      | MS                      |                  | ML                      | CPP                     | DS                      |
| Wed         | CPP                     | PROJECT LAB             |                         |                         |                  | CPP                     | MS                      | ML                      |
| Thu         | MS                      | DS                      | ML                      | LIB                     |                  | DS                      | SEMINAR                 |                         |

Class :IV B.Tech II Semester

w.e.f. 31-03-2021

Sec : D

Class Coordinator :Mr.G.V.Lakshminarayana

Room No :G-402

| Periods     | 1                       | 2                       | 3                       | 4                       | 1:00PM<br>2:00PM | 5                       | 6                       | 7                       |
|-------------|-------------------------|-------------------------|-------------------------|-------------------------|------------------|-------------------------|-------------------------|-------------------------|
| Time<br>Day | (09.30 AM-<br>10.30 AM) | (10.30 AM-<br>11.20 AM) | (11.20 AM-<br>12.10 PM) | (12.10 PM-<br>01.00 PM) |                  | (02.00 PM-<br>02.50 PM) | (02.50 PM-<br>03.40 PM) | (03.40 PM-<br>04.30 PM) |
| Mon         | DS                      | SEMINAR                 |                         | ML                      | LUNCH<br>BREAK   | CPP                     | MS                      | MS                      |
| Tue         | MS                      | CPP                     | DS                      | LIB                     |                  | PROJECT LAB             |                         |                         |
| Wed         | ML                      | MS                      | ML                      | CPP                     |                  | DS                      | DS                      | SPORTS                  |
| Thu         | ML                      | CPP                     | DS                      | SEMINAR                 |                  | MS                      | ML                      | CPP                     |

Head of the Department

## **STAFF DETAILS:**

| <b>S. No.</b> | <b>Course Code</b>   | <b>Course Name</b> | <b>Section</b>          |                                |                                   |                        |
|---------------|----------------------|--------------------|-------------------------|--------------------------------|-----------------------------------|------------------------|
|               |                      |                    | <b>A</b>                | <b>B</b>                       | <b>C</b>                          | <b>D</b>               |
| 1.            | <b>R164205<br/>1</b> | <b>DS</b>          | Mr.M.Baburao            | Mr.M.Baburao                   | Mr.G.V.Lakshminarayana            | Mr.G.V.Lakshminarayana |
| 2.            | <b>R164205<br/>3</b> | <b>ML</b>          | Mr.A.Rajesh             | Mr.A.Rajesh                    | Mr.SheikShabuddin                 | Mr.SheikShabuddin      |
| 3.            | <b>R164205<br/>2</b> | <b>MS</b>          | Ms.B.Aruna              | Mr.P.Bharath Kumar<br>(Course) | Ms.K.LalithaBhavan<br>ni (MODULE) | Ms.K.<br>Muralikrishna |
| 4.            | <b>R164205<br/>A</b> | <b>CPP</b>         | Mrs.D.S.L.Manikateswari | Mrs.D.S.L.Manikateswari        | Ms.D.SasiRekha                    | Ms.D.SasiRekha         |
| 5.            |                      | <b>Seminar</b>     | Mr.M.Baburao            | Mrs. P. Suneetha               | Ms.D.SasiRekha                    | Mr.G.V.Lakshminarayana |
| 6.            |                      | <b>Project Lab</b> | Mr.M.Baburao            | Mrs. P. Suneetha               | Mr.G.V.Lakshminarayana            | Mr.SheikShabuddin      |



Head of the Department

# **COURSE STRUCTURE**

## **IV Year - II Semester**

| <b>S.No</b> | <b>Course</b>                                                                                                               | <b>L</b> | <b>T</b> | <b>P</b> | <b>C</b>  |
|-------------|-----------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|
| 1           | Distributed Systems                                                                                                         | 4        | 0        | 0        | 3         |
| 2           | Management Science                                                                                                          | 4        | 0        | 0        | 3         |
| 3           | Machine Learning                                                                                                            | 4        | 0        | 0        | 3         |
| 4           | <b>Elective-III</b><br>i. Concurrent and Parallel Programming<br>ii. Artificial Neural Networks<br>iii. Operations Research | 4        | 0        | 0        | 3         |
| 5           | Seminar                                                                                                                     | 0        | 3        | 0        | 2         |
| 6           | Project                                                                                                                     | 0        | 0        | 0        | 10        |
|             | <b>Total</b>                                                                                                                | <b>0</b> | <b>0</b> | <b>0</b> | <b>24</b> |

**Total Course Credits = 48+44 + 42 + 46 = 180**



# LESSON PLANS

# **Distributed Systems**

**Academic Year : 2020-21**

**Semester : IV/II**

**Name of the Course: Distributed Systems**

**Programme: B.Tech**

**Sections :A,B,C&D**

**Course Code: 412 / R1642051**

## **COURSE OUTCOMES (Along with Knowledge Level):**

**After completion of this course, the students will be able to:**

| <b>CO</b> | <b>Course Outcome</b>                                                                                          | <b>Knowledge Level</b> |
|-----------|----------------------------------------------------------------------------------------------------------------|------------------------|
| C412.1    | Describe distributed system and desired properties of such systems.                                            | K2                     |
| C412.2    | Discuss the basic theoretical concepts of distributed systems and Characteristics of IPC protocols.            | K2                     |
| C412.3    | Demonstrate a simple distributed system.                                                                       | K3                     |
| C412.4    | Discuss in detail the system level and support required.                                                       | K2                     |
| C412.5    | Explain the mechanisms such as client/server communication, Remote procedure call (RPC/RMI) and P2P algorithms | K2                     |
| C412.6    | Discuss modern distributed systems to meet the demands of contemporary distributed applications                | K2                     |

## **TEXT BOOKS:**

T1. Ajay D Kshemkalyani, Mukesh Sigal, "Distributed Computing, Principles, Algorithms and Systems", Cambridge

T2. George Coulouris, Jean Dollimore, Tim Kindberg, "Distributed Systems- Concepts and Design", Fourth

Edition, Pearson Publication

## **Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):**

| <b>Course Outcome</b> | <b>Targeted Proficiency Level<br/>(% of Marks)</b> | <b>Targeted level of Attainment (%<br/>Students)</b> |
|-----------------------|----------------------------------------------------|------------------------------------------------------|
| 1                     | 60                                                 | 70                                                   |
| 2                     | 60                                                 | 75                                                   |
| 3                     | 60                                                 | 70                                                   |
| 4                     | 60                                                 | 70                                                   |
| 5                     | 60                                                 | 70                                                   |
| 6                     | 60                                                 | 70                                                   |

**CO 1**

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                                          | Knowledge Level of ILO | No. of Hours | Pedagogy                | Teaching aids |
|------|----------------|-------------------------------------------------------------------------------------------|------------------------|--------------|-------------------------|---------------|
| 1    | CO 1           | Dissemination of Vision, Mission, PEOs,POs,PSOs                                           |                        | 1            | Lecture                 | BB            |
| 2    |                | <b>Characterization of Distributed Systems:</b><br>Introduction about Distributed System  | K1                     | 1            | Lecture                 | BB            |
| 3    |                | Explain about examples of Distributed System, Resource Sharing and the Web and Challenges | K2                     | 2            | Lecture with Discussion | ICT           |
| 4    |                | <b>System Models:</b><br>Explain about Architectural Models- Software Layers              | K2                     | 2            | Lecture                 | BB            |
| 5    |                | Describe about System Architecture and Variations                                         | K2                     | 2            | Lecture with Discussion | ICT           |
| 6    |                | Describe about Interface and Objects and Design Requirements for Distributed Architecture | K2                     | 2            | Lecture                 | BB            |
| 7    |                | Discuss about Fundamental Models, Interaction Model                                       | K2                     | 2            | Lecture                 | BB            |
| 8    |                | Discuss about Failure Model and Security Model                                            | K2                     | 1            | Lecture                 | BB            |
|      |                | <b>TOTAL</b>                                                                              |                        | <b>13</b>    |                         |               |

**CO 2**

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                                       | Knowledge Level of ILO | No. of Hours | Pedagogy   | Teaching aids |
|------|----------------|----------------------------------------------------------------------------------------|------------------------|--------------|------------|---------------|
| 1    | CO 2           | <b>Interprocess Communication:</b><br>Introduction The API for the Internet Protocols. | K1                     | 1            | Lecture    | BB            |
| 2    |                | Explain about the Characteristics of Interprocess communication                        | K2                     | 1            | Lecture    | BB            |
| 3    |                | Discuss about Sockets, UDP Datagram Communication                                      | K2                     | 2            | Lecture    | BB            |
| 4    |                | Discuss about TCP Stream Communication                                                 | K2                     | 1            | Discussion | ICT           |
| 5    |                | Explain about External Data Representation and Marshalling                             | K2                     | 1            | Discussion | ICT+BB        |
| 6    |                | Explain about Client Server Communication                                              | K2                     | 2            | Lecture    | BB            |
| 7    |                | Explain about Group Communication                                                      | K2                     | 1            | Lecture    | BB            |
| 8    |                | Discuss about IP Multicast an implementation of group communication,                   | K2                     | 1            | Lecture    | BB            |
| 9    |                | Discuss about Reliability and Ordering of Multicast.                                   | K2                     | 1            | Lecture    | BB            |
|      |                | <b>TOTAL</b>                                                                           |                        | <b>11</b>    |            |               |

**CO 3**

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                       | Knowledge Level of ILO | No. of Hours | Pedagogy                | Teaching aids |
|------|----------------|------------------------------------------------------------------------|------------------------|--------------|-------------------------|---------------|
| 1    | CO 3           | <b>Distributed Objects and Remote Invocation:</b> Introduction         | K1                     | 1            | Lecture                 | BB            |
| 2    |                | Explain about Communication between Distributed Objects-Object Model   | K2                     | 1            | Lecture                 | BB            |
| 3    |                | Discuss about Distributed Object Model                                 | K2                     | 1            | Lecture                 | BB            |
| 4    |                | Discuss about Design Issues for RMI                                    | K2                     | 1            | Discussion With Lecture | ICT           |
| 5    |                | Discuss about Implementation of RMI and Distributed Garbage Collection | K3                     | 2            | Lecture                 | BB            |
| 6    |                | Discuss about Remote Procedure Call                                    | K3                     | 2            | Lecture                 | BB+ICT        |
| 7    |                | Discuss about Events and Notifications                                 | K3                     | 1            | Lecture                 | BB            |
| 8    |                | Review Case Study: JAVA RMI                                            | K2                     | 1            | Lecture                 | BB            |
|      |                | <b>TOTAL</b>                                                           |                        | <b>10</b>    |                         |               |

**CO 4**

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                    | Knowledge Level of ILO | No. of Hours | Pedagogy | Teaching aids |
|------|----------------|---------------------------------------------------------------------|------------------------|--------------|----------|---------------|
| 1    | CO4            | <b>Operating System Support:</b> Introduction                       | K1                     | 1            | Lecture  | BB            |
| 2    |                | Discuss about the Operating System Layer and Protection             | K2                     | 1            | Lecture  | BB            |
| 3    |                | Distinguish between Processes and Threads and Explain Address Space | K2                     | 1            | Lecture  | BB            |
| 4    |                | Explain about Creation of a New Process                             | K2                     | 1            | Lecture  | BB+ICT        |
| 5    |                | Explain about Threads                                               | K2                     | 2            | Lecture  | BB            |
|      |                | <b>TOTAL</b>                                                        |                        | <b>06</b>    |          |               |

**CO 5**

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                                               | Knowledge Level of ILO | No. of Hours | Pedagogy                | Teaching aids |
|------|----------------|------------------------------------------------------------------------------------------------|------------------------|--------------|-------------------------|---------------|
| 1    | CO5            | <b>Distributed File Systems:</b><br>Introduction                                               | K1                     | 1            | Lecture                 | BB            |
| 2    |                | Explain about File Service Architecture                                                        | K2                     | 1            | Lecture                 | BB            |
| 3    |                | Discuss about Peer-to-Peer Systems                                                             | K2                     | 1            | Lecture With Discussion | BB+ICT        |
| 4    |                | Describe about Napster and its Legacy                                                          | K2                     | 1            | Lecture                 | BB            |
| 5    |                | Explain about Peer-to-Peer Middleware                                                          | K2                     | 1            | Discussion              | BB            |
| 6    |                | Explain about Routing Overlays                                                                 | K2                     | 1            | Lecture                 | BB+ICT        |
| 7    |                | <b>Coordination and Agreement:</b><br>Explain about introduction of Coordination and Agreement | K2                     | 1            | Lecture                 | BB            |
| 8    |                | Explain about Distributed Mutual Exclusion                                                     | K2                     | 1            | Lecture                 | BB            |
| 9    |                | Discuss about Elections                                                                        | K2                     | 1            | Lecture                 | BB            |
| 10   |                | Explain about Multicast Communication                                                          | K2                     | 2            | Lecture                 | BB+ICT        |
|      |                |                                                                                                |                        | <b>11</b>    |                         |               |

**CO 6**

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                         | Knowledge Level of ILO | No. of Hours | Pedagogy | Teaching aids |
|------|----------------|--------------------------------------------------------------------------|------------------------|--------------|----------|---------------|
| 1    | CO 6           | <b>Transactions&amp; Replications:</b><br>Introduction                   | K1                     | 1            | Lecture  | BB            |
| 2    |                | Describe about System Model and Group Communication                      | K2                     | 2            | Lecture  | BB            |
| 3    |                | Explain about Concurrency Control in Distributed Transactions,           | K2                     | 1            | Lecture  | BB            |
| 4    |                | Discuss about Distributed Dead Locks                                     | K2                     | 2            | Lecture  | BB+ICT        |
| 5    |                | Explain about Transaction Recovery and introduction of Replication       | K2                     | 2            | Lecture  | BB            |
| 6    |                | Distinguish between Passive (Primary) Replication and Active Replication | K2                     | 1            | Lecture  | BB+ICT        |
|      |                | <b>TOTAL</b>                                                             |                        | <b>09</b>    |          |               |

**Total No. of Classes: 60**

# **Management Science**

**Academic Year : 2020-21**

**Year/ Semester: IV /II**

**Name of the Course: Management Science**

**Programme: B.Tech**

**Section:A, B, C, D**

**Course Code:R1642052**

## **Course Outcomes (Along with knowledge level):**

At the end of the course, student will be able to

|       |                                                                                                                                   |
|-------|-----------------------------------------------------------------------------------------------------------------------------------|
| 411.1 | Clarify various approaches to Management.[K2]                                                                                     |
| 411.2 | Illustrate the principles and practices of operations management.[K3]                                                             |
| 411.3 | Describes the dynamics of individual and interpersonal behavior in organizational setting through human resource management. [K2] |
| 411.4 | Develop the efficient Decision making abilities for the benefit of organization.[K3]                                              |
| 411.5 | Explains the better strategic management for organizational effectiveness.[K2]                                                    |
| 411.6 | Clarify the knowledge of contemporary management practices[K2]                                                                    |

## **TEXT BOOKS SUGGESTED:**

1. Dr. P. Vijaya kumar & Dr. N. Appa rao, management science, cengage.
2. Koontz & wehrich: Essential of Management, TMH 2011

## **REFERENCE BOOKS SUGGESTED:**

1. Philip Kotler & Armstrong: Principles of Management, Pearson publications.
2. Hitt and vijaya kumar: strategic Management, cengage learning.

## **Targeted Proficiency Level (For each course Outcome):**

| <b>Course Outcome</b> | <b>Targeted Proficiency Level</b> |
|-----------------------|-----------------------------------|
| CO1                   | 60                                |
| CO2                   | 60                                |
| CO3                   | 60                                |
| CO4                   | 60                                |
| CO5                   | 60                                |
| CO6                   | 60                                |

**Targeted Level of Attainment (For each course Outcome):**

| <b>Course Outcome</b> | <b>Targeted Attainment Level</b> |
|-----------------------|----------------------------------|
| CO1                   | 65                               |
| CO2                   | 60                               |
| CO3                   | 60                               |
| CO4                   | 60                               |
| CO5                   | 60                               |
| CO6                   | 60                               |

**UNIT - I**

| <b>S.No</b> | <b>Course Outcome</b> | <b>Intended Learning Outcomes (ILO)</b>       | <b>Knowledge Level of ILO</b> | <b>No. of Hours Req'd.</b> | <b>Pedagogy</b>      | <b>Teaching aids</b> |
|-------------|-----------------------|-----------------------------------------------|-------------------------------|----------------------------|----------------------|----------------------|
| 1.          | CO1                   | Definitions of management,                    | K1                            | 1                          | Lecture              | BB                   |
| 2.          |                       | Describe the Functions of management.         | K1                            | 2                          | Lecture              | BB                   |
| 3.          |                       | Evaluation of management thought              | K2                            | 2                          | Lecture+ discussion  | BB                   |
| 4.          |                       | Explain Theories of motivation.               | K2                            | 2                          | Lecture              | BB                   |
| 5.          |                       | Decision-making process                       | K2                            | 1                          | Lecture + discussion | BB                   |
| 6.          |                       | Types of organization structures              | K2                            | 1                          | Lecture + discussion | BB                   |
| 7.          |                       | State the International Management structure. | K2                            | 2                          | Lecture + discussion | BB                   |
|             |                       | <b>Total</b>                                  |                               | <b>11</b>                  |                      |                      |

**UNIT - II**

| <b>S.No</b> | <b>Course Outcome</b> | <b>Intended Learning Outcomes (ILO)</b>                      | <b>Knowledge Level of ILO</b> | <b>No. of Hours Req'd.</b> | <b>Pedagogy</b>      | <b>Teaching aids</b> |
|-------------|-----------------------|--------------------------------------------------------------|-------------------------------|----------------------------|----------------------|----------------------|
| 1.          | CO2                   | Describe the Principles and types of management              | K1                            | 1                          | Lecture              | BB                   |
| 2.          |                       | Work Study and Statistical Quality Control.                  | K2                            | 2                          | Lecture + discussion | BB                   |
| 3.          |                       | Control charts (P- chart, R chart, and C chart)              | K3                            | 2                          | Lecture + discussion | BB                   |
| 4.          |                       | Explain the concept of Material Management                   | K2                            | 1                          | Lecture + discussion | BB                   |
| 5.          |                       | Need for inventory control                                   | K2                            | 2                          | Lecture + discussion | BB                   |
| 6.          |                       | EOQ, ABC analysis simple problems and Types of ABC analysis. | K3                            | 2                          | Lecture + discussion | BB                   |
|             |                       | <b>Total</b>                                                 |                               | <b>10</b>                  |                      |                      |

### UNIT - III

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                       | Knowledge Level of ILO | No. of Hours Reqd. | Pedagogy                                     | Teaching aids |
|------|----------------|--------------------------------------------------------|------------------------|--------------------|----------------------------------------------|---------------|
| 1.   | CO3            | Concept of HRM, HRD and PMIR                           | K1                     | 1                  | Lecture                                      | BB            |
| 2.   |                | Functions of HR manager                                | K2                     | 2                  | Lecture + discussion                         | BB            |
| 3.   |                | Wage payment plans                                     | K2                     | 2                  | Lecture + discussion                         | BB            |
| 4.   |                | Job Evaluation and Merit Rating methods.               | K2                     | 1                  | Lecture + discussion and In-class Assignment | BB+PPT        |
| 5.   |                | Marketing Management based on product life cycle.      | K2                     | 1                  | Lecture + discussion                         | BB            |
| 6.   |                | Channels of Distributions.                             | K2                     | 1                  | Lecture + discussion                         | BB            |
| 7.   |                | Operationlizing change through performance management. | K2                     | 2                  | Lecture + discussion and In-class Assignment | BB+PPT        |
|      |                | <b>Total</b>                                           |                        | <b>10</b>          |                                              |               |

### UNIT - IV

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                                       | Knowledge Level of ILO | No. of Hours Reqd. | Pedagogy                                     | Teaching aids |
|------|----------------|------------------------------------------------------------------------|------------------------|--------------------|----------------------------------------------|---------------|
| 1.   | CO4            | Construction of Network                                                | K2                     | 3                  | Lecture+ discussion                          | BB            |
| 2.   |                | Difference between PERT and CPM                                        | K2                     | 2                  | Lecture + discussion and In-class Assignment | BB+PPT        |
| 3.   |                | Compute Critical path, Probability, Project crashing (Simple problems) | K3                     | 4                  | Lecture + discussion                         | BB            |
|      |                | <b>Total</b>                                                           |                        | <b>09</b>          |                                              |               |

### UNIT - V

| S.No | Course Outcome | Intended Learning Outcomes (ILO)              | Knowledge Level of ILO | No. of Hours Reqd. | Pedagogy                          | Teaching aids |
|------|----------------|-----------------------------------------------|------------------------|--------------------|-----------------------------------|---------------|
| 1.   | C05            | Describe Vision, Mission, Goals and Strategy. | K2                     | 2                  | Lecture + discussion              | BB            |
| 2.   |                | Elements of corporate planning process.       | K2                     | 2                  | Lecture + discussion              | BB            |
| 3.   |                | Environmental Scanning, SWOT analysis.        | K2                     | 2                  | Lecture + discussion and In-class | BB+PPT        |

|    |  |                                                                                              |    |           |                                              |        |
|----|--|----------------------------------------------------------------------------------------------|----|-----------|----------------------------------------------|--------|
|    |  |                                                                                              |    |           | Assignment                                   |        |
| 4. |  | Explain the Steps in Strategy Formulation and Implementation, Generic Strategy alternatives. | K2 | 1         | Lecture                                      | BB     |
| 5. |  | Global Strategies, theories of Multinational Compnies.                                       | K2 | 1         | Lecture + discussion and In-class Assignment | BB+PPT |
|    |  | <b>Total</b>                                                                                 |    | <b>08</b> |                                              |        |

#### UNIT – VI

| S.No | Course Outcome | Intended Learning Outcomes (ILO)                        | Knowledge Level of ILO | No. of Hours Req d. | Pedagogy                                     | Teaching aids |
|------|----------------|---------------------------------------------------------|------------------------|---------------------|----------------------------------------------|---------------|
| 1.   | CO6            | Explain the Concept of MIS                              | K2                     | 1                   | Lecture + discussion                         | BB            |
| 2.   |                | Describe the concept of MRP,                            | K2                     | 2                   | Lecture + discussion                         | BB            |
| 3.   |                | Describe the concept of Just in time.                   | K2                     | 1                   | Lecture + discussion and In-class Assignment | BB+PPT        |
| 4.   |                | Describe the concept of Total Quality Management        | K1                     | 1                   | Lecture                                      | BB            |
| 5.   |                | Describe the concept of Six sigma                       | K2                     | 1                   | Lecture + discussion                         | BB            |
| 6.   |                | Describe the concept of Capability maturity model.      | K1                     | 1                   | Lecture                                      | BB            |
| 7.   |                | Describe the concept of Supply chain Management.        | K2                     | 1                   | Lecture + discussion                         | BB            |
| 8.   |                | Describe the concept of Enterprise Resource planning    | K2                     | 1                   | Lecture + discussion and In-class Assignment | BB+PPT        |
| 9.   |                | Describe the concept of Business process out sourcing   | K2                     | 1                   | Lecture + discussion                         | BB            |
| 10.  |                | Explain the concept of business process Re engineering. | K2                     | 1                   | Lecture + discussion                         | BB            |
| 11.  |                | Describe the concept of Bench marking                   | K2                     | 1                   | Lecture + discussion and In-class Assignment | BB+PPT        |
| 12.  |                | Explain the concept of Balance score card.              | K1                     | 1                   | Lecture                                      | BB            |
|      |                | <b>Total</b>                                            |                        | <b>12</b>           |                                              |               |

**Total No. of classes: 60**

# **Machine Learning**

**Academic Year:** 2020-21

**Programme:** B.Tech

**Year/ Semester:** IV Year II Sem

**Section:** A,B,C,D

**Name of the Course:** MachineLearning

**Course Code:**R1642053

## **Course Outcomes (Along with Knowledge Level):**

Upon completion of the course student will be able to

|   |                                                                                                         |
|---|---------------------------------------------------------------------------------------------------------|
| 1 | Explain the characteristics of machine learning that make it useful to real-world Problems. <b>(K2)</b> |
| 2 | Discuss Machine learning algorithms as supervised, semi-supervised, and Unsupervised.. <b>(K2)</b>      |
| 3 | Demonstrate the Tree and Rule based models <b>(K3)</b>                                                  |
| 4 | Illustrate the Linear and Distance based models <b>(K3)</b>                                             |
| 5 | Demonstrate Regression algorithms.. <b>(K3)</b>                                                         |
| 6 | Apply the Neural networks and Dimensionality reduction for non-linear functions <b>(K3)</b>             |

## **TEXT BOOKS:**

1. Machine Learning: The art and science of algorithms that make sense of data, Peter Flach, Cambridge.
2. Machine Learning, Tom M. Mitchell, MGH.

## **REFERENCE BOOKS:**

1. Understanding Machine Learning: From Theory to Algorithms, Shai Shalev-Shwartz, Shai Ben-David, Cambridge.
2. Machine Learning in Action, Peter Harington, 2012, Cengage.

## **Targeted Proficiency Level (For each course Outcome):**

| <b>Course Outcome</b> | <b>Targeted Proficiency Level</b> |
|-----------------------|-----------------------------------|
| CO1                   | 60                                |
| CO2                   | 60                                |
| CO3                   | 60                                |
| CO4                   | 60                                |
| CO5                   | 60                                |
| CO6                   | 60                                |

### Targeted Level of Attainment (For each course Outcome):

| Course Outcome | Targeted Attainment Level |
|----------------|---------------------------|
| CO1            | 60                        |
| CO2            | 60                        |
| CO3            | 60                        |
| CO4            | 60                        |
| CO5            | 60                        |
| CO6            | 60                        |

### UNIT-I: The ingredients of machine learning, Tasks

| S# | Course Outcomes | Intended Learning Outcomes (ILO)                                                       | Knowledge Level of ILO | No. of Hours Req'd. | Pedagogy                                     | Teaching aids  |
|----|-----------------|----------------------------------------------------------------------------------------|------------------------|---------------------|----------------------------------------------|----------------|
| 1  | CO1             | Dissemination of Course Outcomes.<br>Identify the importance of Machine Learning       | K1                     | 1                   | Lecture                                      | BB             |
| 2  |                 | Illustrate the meaning of the task(problems) that can be solved with machine learning, | K2                     | 2                   | Lecture + discussion and In-class Assignment | BB & Projector |
| 3  |                 | Illustrate the classifying the models in Machine Learning                              | K2                     | 2                   | Lecture + discussion                         | BB & Projector |
| 4  |                 | Illustrate Features, and the workhorses of machine Learning.                           | K2                     | 2                   | Lecture + discussion                         | BB& Projector  |
| 5  |                 | Illustrate Classification, Scoring and ranking                                         | K2                     | 2                   | Lecture + discussion                         | BB& Projector  |
|    |                 | Total                                                                                  |                        | 9                   |                                              |                |

### UNIT-II: Beyond binary classification

| S# | Course Outcomes | Intended Learning Outcomes (ILO)                  | Knowledge Level of ILO | No. of Hours Required | Pedagogy                | Teaching aids  |
|----|-----------------|---------------------------------------------------|------------------------|-----------------------|-------------------------|----------------|
| 1  | CO2             | Demonstrate Handling more than two classes        | K2                     | 2                     | Lecture + Demonstration | BB & Projector |
| 2  |                 | Demonstrate Regression,                           | K2                     | 2                     | Lecture + Demonstration | BB & Projector |
| 3  |                 | Illustrate Unsupervised and descriptive learning. | K2                     | 1                     | Lecture + Demonstration | BB & Projector |
| 4  |                 | Demonstrate The hypothesis space                  | K2                     | 2                     | Lecture + Demonstration | BB & Projector |

|   |  |                                                 |    |   |                         |                |
|---|--|-------------------------------------------------|----|---|-------------------------|----------------|
| 5 |  | Demonstrate Paths through the hypothesis space, | K2 | 1 | Lecture + Demonstration | BB & Projector |
| 6 |  | Demonstrate Beyond conjunctive concepts         | K2 | 1 | Lecture + Demonstration | BB & Projector |
|   |  | Total                                           |    | 9 |                         |                |

#### Unit-III: Tree models and Rule models

| S# | Course Outcomes | Intended Learning Outcomes (ILO)                      | Knowledge Level of ILO | No. of Hours Required | Pedagogy                | Teaching aids  |
|----|-----------------|-------------------------------------------------------|------------------------|-----------------------|-------------------------|----------------|
| 1  | CO3             | Demonstrate Decision trees                            | K3                     | 2                     | Lecture + Demonstration | BB & Projector |
| 2  |                 | Demonstrate Ranking and probability estimation trees, | K3                     | 2                     | Lecture + Demonstration | BB & Projector |
| 3  |                 | Demonstrate Tree learning as variance reduction.      | K3                     | 2                     | Lecture + Demonstration | BB & Projector |
| 4  |                 | Demonstrate Learning ordered rule lists,              | K3                     | 2                     | Lecture + Demonstration | BB & Projector |
| 5  |                 | Demonstrate Learning unordered rule sets              | K3                     | 2                     | Lecture + Demonstration | BB & Projector |
| 6  |                 | Demonstrate Descriptive rule learning                 | K3                     | 2                     | Lecture + Demonstration | BB & Projector |
| 7  |                 | Demonstrate First-order rule learning                 | K3                     | 2                     | Lecture + Demonstration | BB & Projector |
|    |                 | Total                                                 |                        | 14                    |                         |                |

#### Unit-IV: Linear models and Distance Based Models

| S# | Course Outcomes | Intended Learning Outcomes (ILO)                                                  | Knowledge Level of ILO | No. of Hours Required | Pedagogy                | Teaching aids  |
|----|-----------------|-----------------------------------------------------------------------------------|------------------------|-----------------------|-------------------------|----------------|
| 1  | CO4             | Illustrate The least-squares method                                               | K2                     | 1                     | Lecture + Demonstration | BB & Projector |
| 2  |                 | Illustrate The perceptron: a heuristic learning algorithm for linear classifiers, | K2                     | 1                     | Lecture + Demonstration | BB & Projector |
| 3  |                 | Demonstrate Support vector machines                                               | K3                     | 2                     | Lecture + Demonstration | BB & Projector |
| 4  |                 | obtaining probabilities                                                           | K3                     | 2                     | Lecture +               | BB &           |

|    |  |                                                       |    |    |                         |                |
|----|--|-------------------------------------------------------|----|----|-------------------------|----------------|
|    |  | from linear classifiers,                              |    |    | Demonstration           | Projector      |
| 5  |  | Illustrate Going beyond linearity with kernel methods | K2 | 1  | Lecture + Demonstration | BB & Projector |
| 6  |  | Illustrate Distance Based Models: Introduction,       | K2 | 1  | Lecture + Demonstration | BB & Projector |
| 7  |  | Demonstrate Neighbors and exemplars,                  | K3 | 2  | Lecture + Demonstration | BB & Projector |
| 8  |  | Demonstrate Nearest Neighbours classification         | K3 | 2  | Lecture + Demonstration | BB & Projector |
| 9  |  | Demonstrate Distance Based Clustering,                | K3 | 2  | Lecture + Demonstration | BB & Projector |
| 10 |  | Demonstrate Hierarchical Clustering.                  | K3 | 2  | Lecture + Demonstration | BB & Projector |
|    |  | Total                                                 |    | 16 |                         |                |

#### UNIT-V: Probabilistic models, Features and Model ensembles

| S# | Course Outcomes | Intended Learning Outcomes (ILO)                                                                                    | Knowledge Level of ILO | No. of Hours Required | Pedagogy                | Teaching aids  |
|----|-----------------|---------------------------------------------------------------------------------------------------------------------|------------------------|-----------------------|-------------------------|----------------|
| 1  | CO5             | Demonstrate The normal distribution and its geometric interpretations,                                              | K3                     | 2                     | Lecture + Demonstration | BB & Projector |
| 2  |                 | Demonstrate Probabilistic models for categorical data,                                                              | K3                     | 2                     | Lecture + Demonstration | BB & Projector |
| 3  |                 | Demonstrate Discriminative learning by optimizing conditional likelihood Probabilistic models with hidden variables | K3                     | 2                     | Lecture + Demonstration | BB & Projector |
| 4  |                 | Demonstrate Kinds of feature, Feature transformations, Feature construction and selection.                          | K3                     | 3                     | Lecture + Demonstration | BB & Projector |
| 5  |                 | Demonstrate Model ensembles: Bagging and random forests, Boosting                                                   | K3                     | 2                     | Lecture + Demonstration | BB & Projector |
|    |                 | Total                                                                                                               |                        | 11                    |                         |                |

**UNIT-VI: Dimensionality Reduction and Artificial Neural Networks**

| S# | Course Outcomes | Contents                                                                                     | Knowledge Level of ILO | No. of Hours Required | Pedagogy                | Teaching aids  |
|----|-----------------|----------------------------------------------------------------------------------------------|------------------------|-----------------------|-------------------------|----------------|
| 1  | CO6             | Principal Component Analysis (PCA), Implementation and demonstration                         | K3                     | 3                     | Lecture + Demonstration | BB & Projector |
| 2  |                 | Illustrate to Introduction to ANN                                                            | K2                     | 2                     | Lecture + Demonstration | BB & Projector |
| 3  |                 | Demonstration Neural network representation appropriate problems for neural network learning | K3                     | 3                     | Lecture + Demonstration | BB & Projector |
|    |                 | Total                                                                                        |                        | 8                     |                         |                |

**Total Hours= 68**

# **Concurrent and Parallel Programming**

Academic Year: 2020-21

Programme: B.Tech

Year/ Semester: IV/II

Section: A, B,C & D

Course Name: Concurrent and Parallel Programming

Course Code: C- 412

## **COURSE OUTCOMES (Along with Knowledge Level):**

**After completion of this course, the students will be able to:**

- CO1. Illustrate concurrent and parallel programming. (K2)
- CO2. Describe issues Process and threads. (K2)
- CO3. Develop Different parallel algorithms. (K3)
- CO4. Describe parallel programming Models. (K2)
- CO5. Illustrate the multiprocessing and multi-threading programming interfaces.(K2)
- CO6. Discuss the Heterogeneous computing techniques.(K2)

## **Text Books suggested:**

- T1. Mordechai Ben-Ari. Principles of Concurrent and Distributed Programming, Prentice-Hall International.
- T 2. Greg Andrews. Concurrent Programming: Principles and Practice, Addison Wesley.
- T 3. GadiTaubenfeld. Synchronization Algorithms and Concurrent Programming, Pearson.
- T 4. M. Ben-Ari. Principles of Concurrent Programming, Prentice Hall.
- T5. Fred B. Schneider. On Concurrent Programming, Springer.
- T 6. Brinch Hansen. The Origins of Concurrent Programming, From Semaphore.

## **Reference Books suggested:**

- R1: Matthew J. Sottile, Introduction to Concurrency in Programming Languages, CRC Press
- R2: Joe Duffy, Concurrent Programming on Windows

**Targeted Proficiency Level and Targeted level of Attainment (for each Course Outcome):**

| <b>Course Outcome</b> | <b>Targeted Proficiency Level<br/>(% of Marks)</b> | <b>Targeted level of Attainment<br/>(% Students)</b> |
|-----------------------|----------------------------------------------------|------------------------------------------------------|
| C- 412.1              | 60                                                 | 60                                                   |
| C- 412.2              | 60                                                 | 60                                                   |
| C- 412.3              | 60                                                 | 60                                                   |
| C- 412.4              | 60                                                 | 60                                                   |
| C- 412.5              | 60                                                 | 60                                                   |
| C- 412.6              | 60                                                 | 60                                                   |

**Unit-1**

| <b>S.No</b> | <b>Course Outcome</b> | <b>Intended Learning Outcomes<br/>(ILO)</b>               | <b>Knowledge Level of ILO</b> | <b>No. of Hours</b> | <b>Pedagogy</b> | <b>Teaching aids</b> |
|-------------|-----------------------|-----------------------------------------------------------|-------------------------------|---------------------|-----------------|----------------------|
| 1           | <b>CO1</b>            | Generalize Concurrent Programming sequential programming. | K2                            | 1                   | Lecture         | BB                   |
|             |                       | Explain Concurrent versus sequential programming.         | K2                            | 1                   |                 |                      |
| 2           |                       | Discuss Concurrent programming constructs.                | K2                            | 2                   | Lecture         | BB                   |
| 3           |                       | Clarify Concurrent programming race condition.            | K2                            | 2                   | Lecture         | BB                   |
| 4           |                       | Describe Synchronization primitives.                      | K2                            | 2                   | Lecture         | BB+ICT               |

**Unit- 2**

| <b>S.No</b> | <b>Course Outcome</b> | <b>Intended Learning Outcomes<br/>(ILO)</b> | <b>Knowledge Level of ILO</b> | <b>No. of Hours</b> | <b>Pedagogy</b>         | <b>Teaching aids</b> |
|-------------|-----------------------|---------------------------------------------|-------------------------------|---------------------|-------------------------|----------------------|
| 1           | <b>CO2</b>            | Classify Processes and threads.             | K2                            | 2                   | Lecture                 | BB                   |
| 2           |                       | Explain Interprocess communication.         | K2                            | 2                   | Lecture with Discussion | BB                   |
| 3           |                       | Generalize Live lock and deadlocks.         | K2                            | 2                   | Lecture with Discussion | BB+ICT               |

|   |  |                                                                  |    |   |                         |        |
|---|--|------------------------------------------------------------------|----|---|-------------------------|--------|
| 4 |  | Clarify starvation, and deadlock prevention.                     | K2 | 2 | Lecture                 | BB     |
| 5 |  | Review Issues and challenges in concurrent programming paradigm. | K2 | 2 | Lecture with Discussion | BB+ICT |
| 6 |  | Expalin concurrent programming paradigm current trends.          | K2 | 2 | Lecture with Discussion | BB+ICT |

### Unit-3

| S.No | Course Outcome | Intended Learning Outcomes (ILO)               | Knowledge Level of ILO | No. of Hours | Pedagogy                | Teaching aids |
|------|----------------|------------------------------------------------|------------------------|--------------|-------------------------|---------------|
| 1    | CO3            | Apply Parallel algorithms for sorting.         | K3                     | 2            | Lecture                 | BB            |
| 2    |                | Find Parallel algorithms for ranking.          | K3                     | 2            | Lecture with Discussion | BB + ICT      |
| 3    |                | Illustrate Parallel algorithms for searching.  | K3                     | 2            | Lecture with Discussion | BB + ICT      |
| 4    |                | Apply Parallel algorithms for traversals.      | K3                     | 2            | Lecture with Discussion | BB+ ICT       |
| 5    |                | Illustrate Parallel algorithms for prefix sum. | K3                     | 2            | Lecture with Discussion | BB+ ICT       |

### Unit- 4

| S.No | Course Outcome | Intended Learning Outcomes (ILO)          | Knowledge Level of ILO | No. of Hours | Pedagogy                | Teaching aids |
|------|----------------|-------------------------------------------|------------------------|--------------|-------------------------|---------------|
| 1    | CO4            | Discuss Parallel programming paradigms.   | K2                     | 1            | Lecture                 | BB            |
|      |                | Classify Data parallel.                   | K2                     | 1            | Lecture                 | BB            |
| 2    |                | Explain Task parallel.                    | K2                     | 2            | Lecture with Discussion | BB+ ICT       |
| 3    |                | Extend Shared memory and message passing. | K2                     | 2            | Lecture with            | BB+ ICT       |

|   |  |                                  |    |   |                         |         |
|---|--|----------------------------------|----|---|-------------------------|---------|
|   |  |                                  |    |   | Discussion              |         |
| 4 |  | Estimate Parallel Architectures. | K2 | 2 | Lecture with Discussion | BB+ ICT |
| 5 |  | Explain GPGPU.                   | K2 | 2 | Lecture with Discussion | BB+ ICT |
| 6 |  | Clarify pthreads.                | K2 | 2 | Lecture with Discussion | BB+ ICT |
| 7 |  | Discuss STM.                     | K2 | 2 | Lecture                 | BB      |

### Unit-5

| S.No | Course Outcome | Intended Learning Outcomes (ILO) | Knowledge Level of ILO | No. of Hours | Pedagogy                | Teaching aids |
|------|----------------|----------------------------------|------------------------|--------------|-------------------------|---------------|
| 1    | CO5            | Illustrate Open Mp.              | K2                     | 2            | Lecture with Discussion | BB            |
| 2    |                | Explain OpenCL.                  | K2                     | 2            | Lecture with Discussion | BB + ICT      |
| 3    |                | Express Cilk++.                  | K2                     | 2            | Lecture with Discussion | BB + ICT      |
| 4    |                | Illustrate Intel TBB.            | K2                     | 2            | Lecture with Discussion | BB + ICT      |
| 5    |                | Describe CUDA.                   | K2                     | 2            | Lecture with Discussion | BB + ICT      |

**Unit-6**

| S.No | Course Outcome | Intended Learning Outcomes (ILO)   | Knowledge Level of ILO | No. of Hours | Pedagogy                | Teaching aids |
|------|----------------|------------------------------------|------------------------|--------------|-------------------------|---------------|
| 1    | CO6            | Associate Heterogeneous Computing. | K2                     | 2            | Lecture with practical  | BB + ICT      |
| 2    |                | Express C++AMP.                    | K2                     | 2            | Lecture with practical  | BB + ICT      |
| 3    |                | Describe Open CL.                  | K2                     | 2            | Lecture with Discussion | BB + ICT      |

**Total No. of Classes: 60**